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MULTIVARIABLE CALCULUS • UNIT 6.01

Vector Fields, Conservative Fields & Line Integrals

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Vector Fields, Conservative Fields & Line Integrals Practice Exercises

1 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

A $\langle 6, 8 \rangle$

B $\langle 8, 6 \rangle$

C $\langle 5, 6 \rangle$

D $\langle 6, 5 \rangle$

2 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 2x, 3y \rangle$. Which of the following options best represents the vector field at the point (1, 1)?

A $\langle 2, 3 \rangle$

B $\langle 1, 1 \rangle$

C $\langle 4, 6 \rangle$

D $\langle 2, 6 \rangle$

3 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$. What is the value of the vector field at the point $(3, -1)$?

A $\langle 5, 6 \rangle$

B $\langle 5, 0 \rangle$

C $\langle 7, 6 \rangle$

D $\langle 7, 0 \rangle$

4 MCQ

Evaluate $\int_C x^2 \, ds$ where C is the segment from $(0,\ 0)$ to $(5,\ 12)$.

A $\frac{25}{3}$

B $\frac{4225}{3}$

C $\frac{328}{3}$

D $\frac{325}{3}$

5 MCQ

Which vector field is conservative on all of R^2?

A $\langle e^y, xe^y \rangle$

B $\langle xy, x \rangle$

C $\langle x + y, -x + y \rangle$

D $\langle x^2 y, xy^2 \rangle$

6 MCQ

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y^2, 2x \rangle$ along the segment from $(-2,\ -2)$ to $(-2,\ 3)$.

A 20

B -19

C -40

D -20

7 MCQ

Evaluate $\int_C x + y\, ds$ where C is the segment from $(0,\ 0)$ to $(4,\ 3)$.

A $\frac{35}{2}$

B $\frac{7}{2}$

C $\frac{175}{2}$

D $\frac{37}{2}$

8 MCQ

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle xy, y \rangle$ along the segment from $(1,\ 1)$ to $(1,\ -2)$.

A $-\frac{3}{2}$

B $\frac{5}{2}$

C $\frac{3}{2}$

D 3

9 MCQ

$\mathbf{F} = \langle ye^x, e^x \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(2,\ 0)$ to $(1,\ -1)$.

A $1 - e$

B $-2e$

C $-e$

D e

10 MCQ

$\mathbf{F} = \langle y^2 + 1, 2xy \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0,\ 0)$ to $(-2,\ 3)$.

A -20

B 20

C -19

D -40

11 MCQ

$\mathbf{F} = \langle y, x + z, y \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ 2,\ 0)$ to $(-1,\ -1,\ 3)$.

A -4

B 4

C -2

D -3

12 MCQ

$\mathbf{F} = \langle 2x + y, x + 3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-1,\ 0)$ to $(0,\ -2)$.

A -8

B -18

C -9

D 9

13 MCQ

$\mathbf{F} = \langle 2xy, x^2 - 2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-2,\ 0)$ to $(1,\ 3)$.

A -3

B 3

C -2

D -6

14 MCQ

$\mathbf{F} = \langle 2x + y, x + 3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-2,\ 0)$ to $(0,\ 0)$.

A 4

B -4

C 0

D -3

15 MCQ

$\mathbf{F} = \langle 2x + 3y, 3x + 3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ -1)$ to $(2,\ -2)$.

A -16

B -13

C -12

D 13

16 MCQ

$\mathbf{F} = \langle y, x + z, y \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ 0,\ 0)$ to $(-2,\ 2,\ 3)$.

A -2

B 3

C 2

D 4

17 MCQ

$\mathbf{F} = \langle y^2 + 1, 2xy \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ 0)$ to $(2,\ 3)$.

A -19

B 19

C 20

D 38

18 MCQ

$\mathbf{F} = \langle 2x + 2y, 2x + 3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1, -1)$ to $(-2, 1)$.

A -3

B 1

C 3

D 4

19 MCQ

$\mathbf{F} = \langle y\cos(x) + 1, \sin(x) \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-2, 1)$ to $(0, -2)$.

A $\sin(2) + 2$

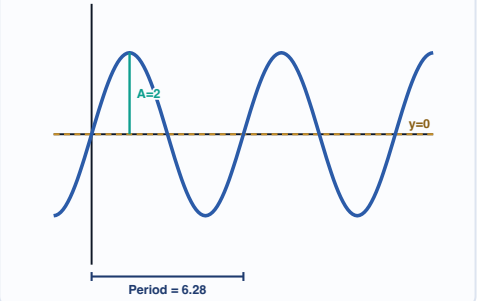
B $-2 - \sin(2)$

C 0

D $\sin(2) + 3$

20 MCQ

$\mathbf{F} = \langle y\cos(x) + 1, \sin(x) \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-2, -1)$ to $(-2, 1)$.



A $2\sin(2)$

B $-2 - \sin(2)$

C $1 - 2\sin(2)$

D $-2\sin(2)$

21 MCQ

$\mathbf{F} = \langle 2x + 3y, 3x + 3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1, 2)$ to $(-1, 0)$.

A -14

B 14

C 1

D -13

22 MCQ

$\mathbf{F} = \langle y^2 + 1, 2xy \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1, -1)$ to $(3, 2)$.

A 14

B 13

C -13

D 15

23 MCQ

$\mathbf{F} = \langle 2xy, x^2 - 2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ 0)$ to $(2,\ -2)$.

A 4

B -4

C -3

D -8

24 MCQ

$\mathbf{F} = \langle y, x + z, y \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0,\ -2,\ 1)$ to $(-2,\ 0,\ 2)$.

A -2

B 0

C 3

D 2

25 MCQ

$\mathbf{F} = \langle y\cos(x) + 1, \sin(x) \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0,\ 2)$ to $(-2,\ 3)$.

A $2 + 3\sin(2)$

B $-3\sin(2) - 1$

C $-6\sin(2) - 4$

D $-3\sin(2) - 2$